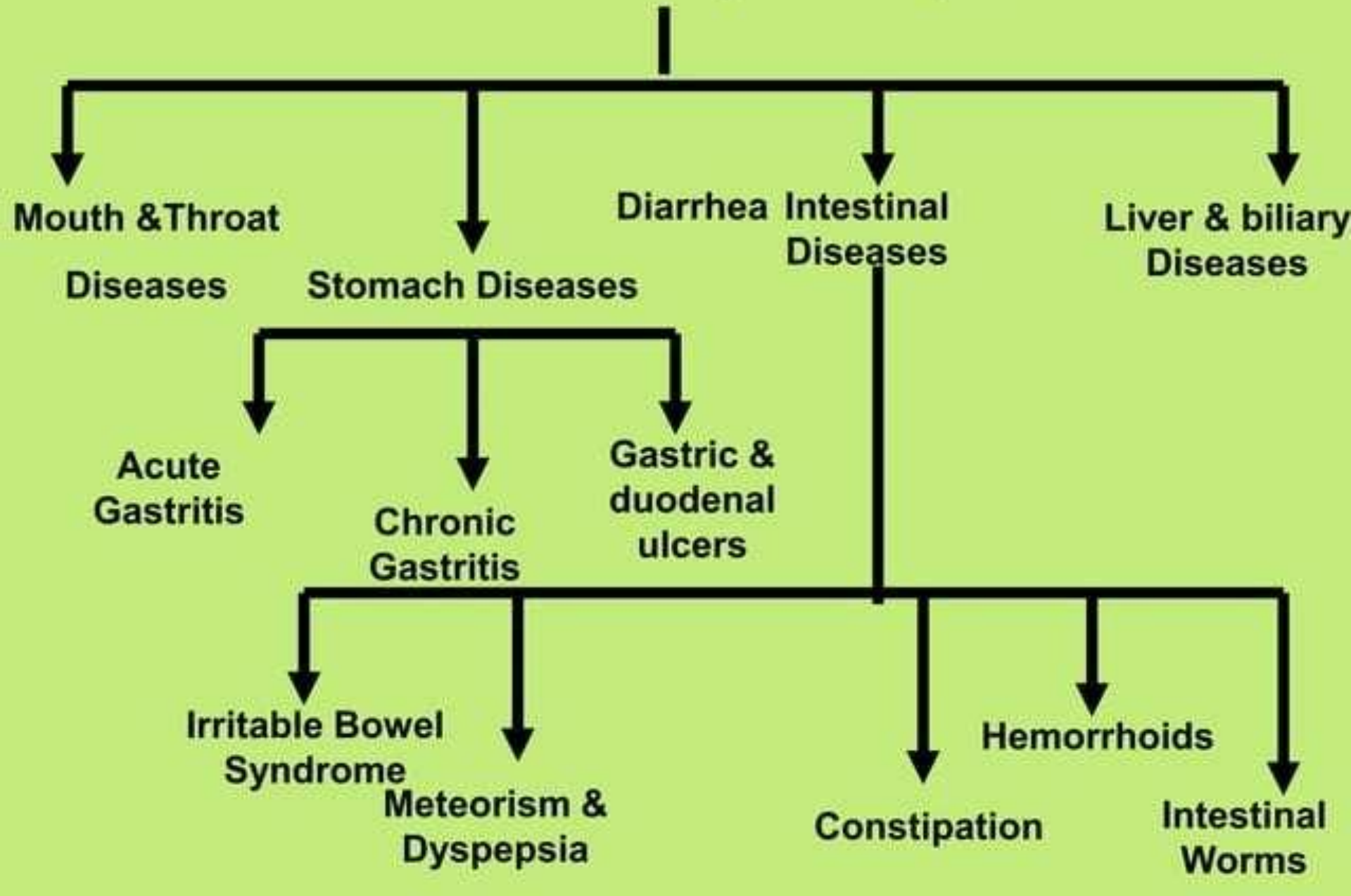
Digestive system

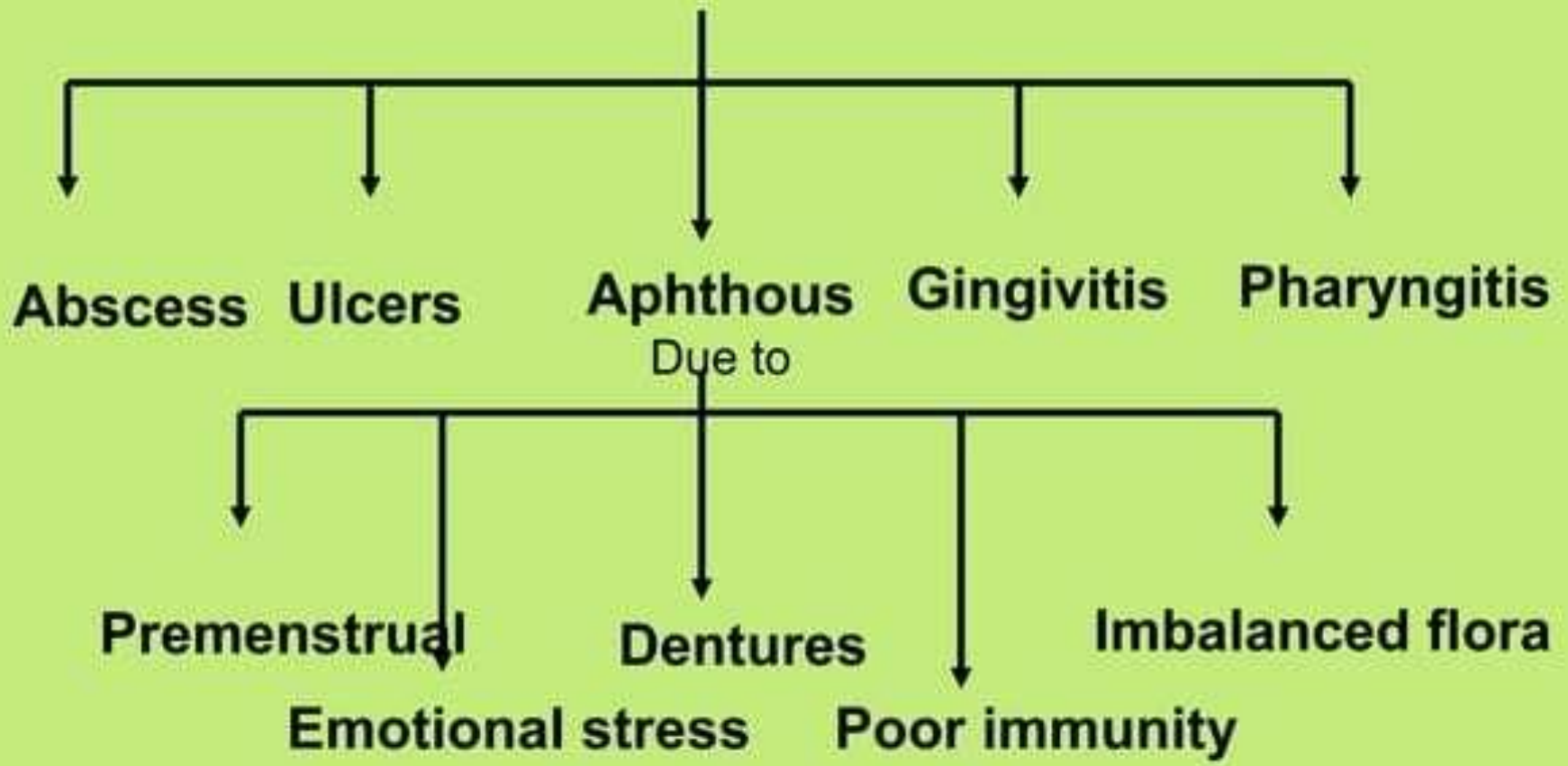
- Herbal medicine is indicated in a wide range of gastrointestinal and metabolic diseases and may be the drugs of choice in these indications.

- These gastrointestinal diseases can be classified as the following

Diseases of the digestive system



I-Mouth disorders



I- Mouth and throat disease

The herbs used in mouth and throat washes are divided into four categories according to their primary actions, namely

1- Demulcent

2-Topical antiphlogistic

3-Astringent

4- Tonic.

- Mouthwashes, gargles and lozenges are still the most effective and reliable way to treat various diseases of the mouth and pharynx.
- They are useful in inflammations ranging from ulcers, peritonsillar abscess, stomatitis, and gingivitis to acute and chronic pharyngitis.

Classification of herbal remedies used in Mouth and throat disease

1. Demulcents (Mucilage-Rich Herbs)

- They form a protective and soothing layer over the mucous membranes, relieving irritation and inflammation.
- **Examples:**

  Marshmallow, Slippery Elm, Plantago (Plantain)
- **Mechanism of Action:**
- Mucilage in these herbs forms a protective coating over inflamed tissues, reducing pain and irritation.
- They retain moisture, preventing dryness of the throat and promoting healing.
- **Clinical Uses:** Helpful in pharyngitis, tonsillitis, mouth ulcers, and dry mouth conditions

- **2. Topical Anti-Inflammatory Herbs**
- These herbs reduce inflammation, swelling, and redness of the mucosa, aiding in faster healing.
- **Examples:**
- Chamomile→ The primary herbal anti-inflammatory agent.
- Arnica flowers → Stimulates local circulation, improving mucosal defenses.
- Sage leaves (Salvia officinalis) → Has antimicrobial and anti-inflammatory properties.
- **Clinical Uses:**
- Chamomile and sage gargles help with sore throat, gingivitis, and ulcerative conditions.
- Arnica is primarily used externally (not for internal use in mouthwashes) for oral mucosal injuries.

- **3. Astringents (Tannin-Containing Herbs)**
- Astringents help tighten and tone tissues, reducing excess secretions and providing relief in inflammatory conditions.
- **Examples:** Witch hazel (Hamamelis virginiana, Krameria (Rhatany) and Catechu
- **Mechanism of Action:**
- Tannins in these plants precipitate proteins, leading to tissue contraction and reduction in inflammation.
- They create a protective barrier that limits bacterial penetration and soothes ulcers.
- **Clinical Uses:**
- Used in mouth rinses and gargles for bleeding gums, canker sores, and pharyngitis.
- Often included in herbal toothpaste and oral care products.

- **4.Tonics (Bitters)**
- Bitter herbs stimulate digestive secretions and enhance overall mucosal function.
- **Examples:** Gentian, Centaurium and Quassia
- Mechanism of Action:
- Stimulate taste receptors, which in turn increase salivary and gastric secretions.
- Improve mucosal blood flow, supporting tissue regeneration.
- **Clinical Uses:**
- Useful in chronic stomatitis and conditions where the oral mucosa is inactive or atrophied.
- Gargling with bitters stimulates appetite and is beneficial in conditions where low gastric acid secretion is an issue.

Echinacea

- Echinacea, a popular medicinal herb often used to support immune health, contains various active constituents

1-Alkamides:

- These are thought to contribute to echinacea's immune-modulating effects, influencing cytokine production and providing anti-inflammatory effects.

2-Polysaccharides:

- Complex sugars like arabinogalactan are believed to stimulate immune cell activity, such as macrophages.



3-Caffeic Acid Derivatives:

- echinacoside and cichoric acid, which have antioxidant properties and may support immune function

Echinacea

Role in Wound Healing & Ulcer Repair

- Inhibiting Hyaluronidase Enzyme which breaks down hyaluronic acid, an essential component of connective tissues.
- By inhibiting this enzyme, Echinacea preserves tissue integrity, promoting faster wound healing.

Topical Benefits for Oral & Skin Ulcers

- Clinical evidence supports Echinacea-based ointments and mouthwashes for treating ulcers, gingivitis, and pharyngitis.

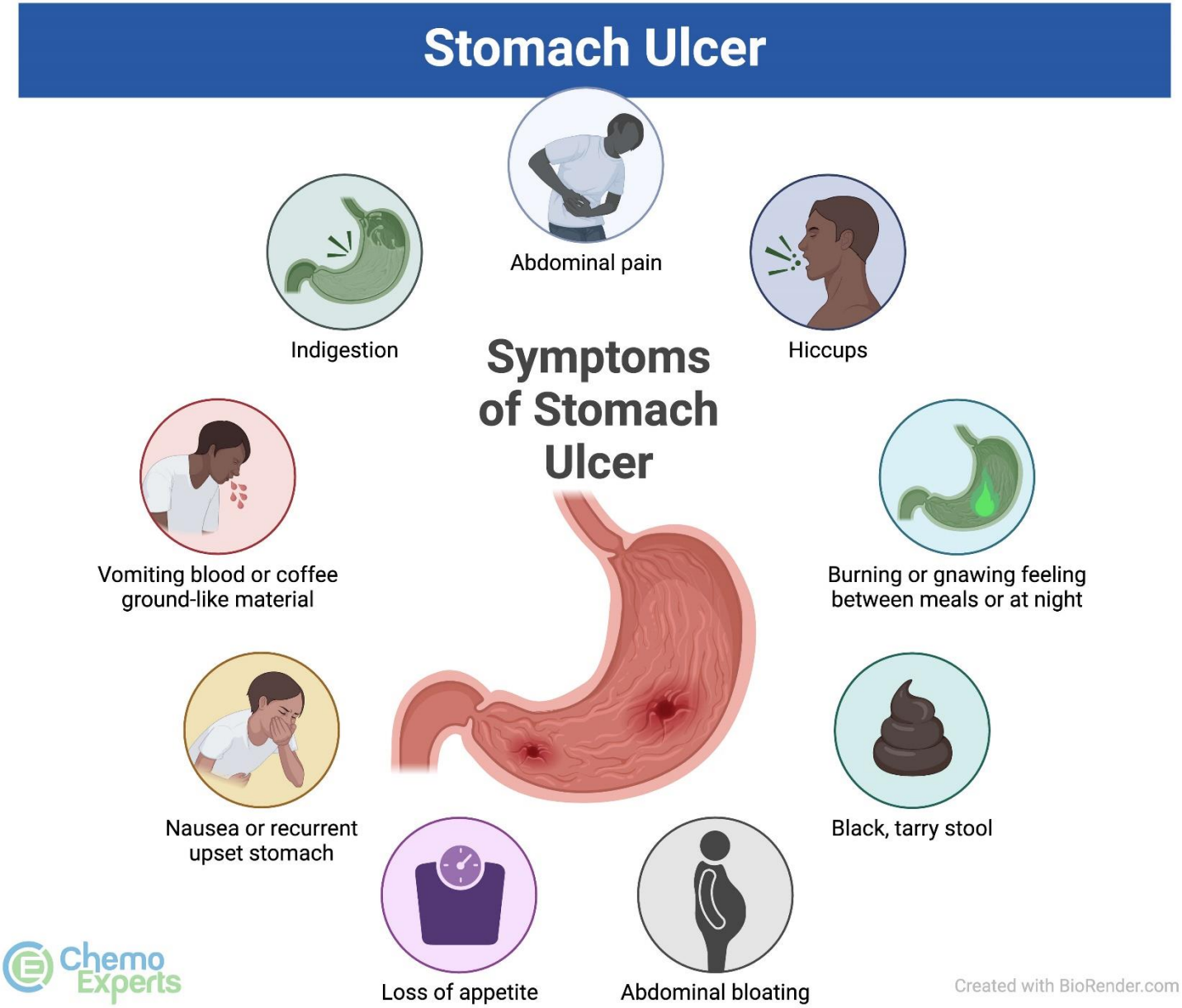
Peptic ulcers

- Peptic ulcers are open sores or erosions in the lining of either the duodenum (duodenal ulcers) or the stomach (gastric ulcers).



- About 10% of all Americans get ulcers, and they can recur.
- Contrary to popular belief, ulcers are not caused by spicy food or stress but, rather, are most commonly due to either an infection or long term use of certain medications.

Signs and Symptoms



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CAUSES OF PEPTIC ULCERS



ETIOLOGY/ RISK FACTORS

- Lifestyle
  - Smoking
  - Acidic drinks
  - Medications
- H. Pylori infection
  - 90% have this bacterium
  - Passed from person to person (fecal-oral route or oral-oral route)
- Age
  - Duodenal 30-40
  - Gastric over 50
- Gender
  - Duodenal: are increasing in older women
- Genetic factors
  - More likely if family member has Hx
- Other factors: stress can worsen but not the cause

Diagnostic Tests

NON-INVASIVE TEST

- ***^{13}C Urea Breath test*** *- Pt. with or without hx of prior peptic ulcer disease*
- ***H. pylori* Stool Antigen (HpSA) -** Pt. with or without hx of prior peptic ulcer disease
- **Serology** – *H. pylori* antibody titers
  - *Used for pt. with **NO** hx of prior H. pylori infection*

INVASIVE TEST

- **Endoscopy** – Biopsy and histological study
  - *Pt. with or without hx of prior peptic ulcer disease*

Treatment Approach

The main goals for treating a peptic ulcer include

1-Eliminating the underlying cause (particularly *H. pylori* infection or use of NSAIDs).

2-Preventing further damage and complications

3-Reducing the risk of recurrence

Life style

- Doctors used to recommend eating bland foods with milk and only small amounts of food with each meal.
- Eat a diet rich in fiber, especially from fruits and vegetables
- Foods containing flavonoids, like apples, celery, cranberries (including cranberry juice), onions, and tea may inhibit the growth of *H. pylori*.
- Quit smoking and alcohol
- Cut down on coffee, as well as carbonated beverages all of which can increase stomach acid
- Reduce stress

Medications

Category	Subcategory	Examples
Reduction of gastric acid secretion	H_2 antihistamines	Cimetidine, ranitidine, famotidine, roxatidine
	PPIs	Omeprazole, lansoprazole, pantoprazole, rabeprazole, dexrabeprazole
	Anticholinergic drugs	Pirenzepine, propantheline, oxyphenomium
	PG analog	Misoprostol
Antacids	Systemic	Sodium bicarbonate, sodium citrate
	Nonsystemic	Magnesium hydroxide, magnesium trisilicate, aluminum hydroxide, calcium carbonate
Ulcer protectives	-	Sucralfate, colloidal bismuth subcitrate
Ulcer-healing drug	-	Carbenoxolone sodium
Anti-*H. pylori* drug	-	Amoxicillin, metronidazole, tinidazole, tetracycline

PPIs: Proton pump inhibitors, *H. pylori*: *Helicobacter pylori*, PG: Prostaglandin

Nutrition and Dietary Supplements

1-Gamma-linolenic acid (GLA):

gamma-linolenic acid (GLA) from evening primrose oil (EPO) may have anti-ulcer properties.

2-Probiotics:

Healthy or "friendly" organisms, called probiotics, inhabit the lining of the intestines and protect us from the entrance of "bad" infections that can cause disease.

Lactobacillus acidophilus (L. acidophilus) is the most commonly used probiotic that were able to kill or slow down the growth of H. pylori

Herbal remedies

1-Astragalus:

Used traditionally to treat stomach ulcers.

Anti-inflammatory → Reduces gastric irritation by inhibiting pro-inflammatory cytokines.

Immunomodulatory → Enhances immune response, which may help combat H. pylori infections.

2-Barberry:

This herb berberine alkaloids that have been shown to combat infection and bacteria. For this reason, barberry is used to ease inflammation and infection of the gastrointestinal tract.

Herbal remedies

3-Bilberry:

Anthocyanidins (an antioxidant) from bilberry fruits help prevent stomach ulcers related to a variety of factors including stress, medications, and alcohol.

4-Cat's Claw

The bark and root of this herb have been for centuries to treat ulcers and other gastrointestinal disorders.

The benefits of this herb may be due to its ability to reduce inflammation. Contains oxindole alkaloids that reduce gastric inflammation.

Herbal remedies

5-Liqourice (*Glycyrrhiza glabra*)

.This herb is a demulcent that has long been valued for its use in food and medicinal remedies, including treatment of ulcers.

Mechanism of Action:

•Demulcent (Coating Effect) → Forms a protective layer over stomach lining.

•Stimulates Mucus Production → Enhances gastric defense mechanisms.

•Supports Healing of Ulcers → Promotes epithelial regeneration.

.Some Liqourice root extracts, known as deglycyrrhizinated Liqourice (DGL), still have the healing properties of licorice without the harmful effects (like high blood pressure).

Herbal remedies

6-Turmeric (*Curcuma longa*)

- Extracts of turmeric root reduced the release of acid from the stomach and protected against injuries such as gastritis or inflammation of the intestinal walls and ulcers.
- **Note:**
- At very high doses, turmeric may induce ulcers.
- It is very important to stick with the dose recommended by an herbal specialist.)

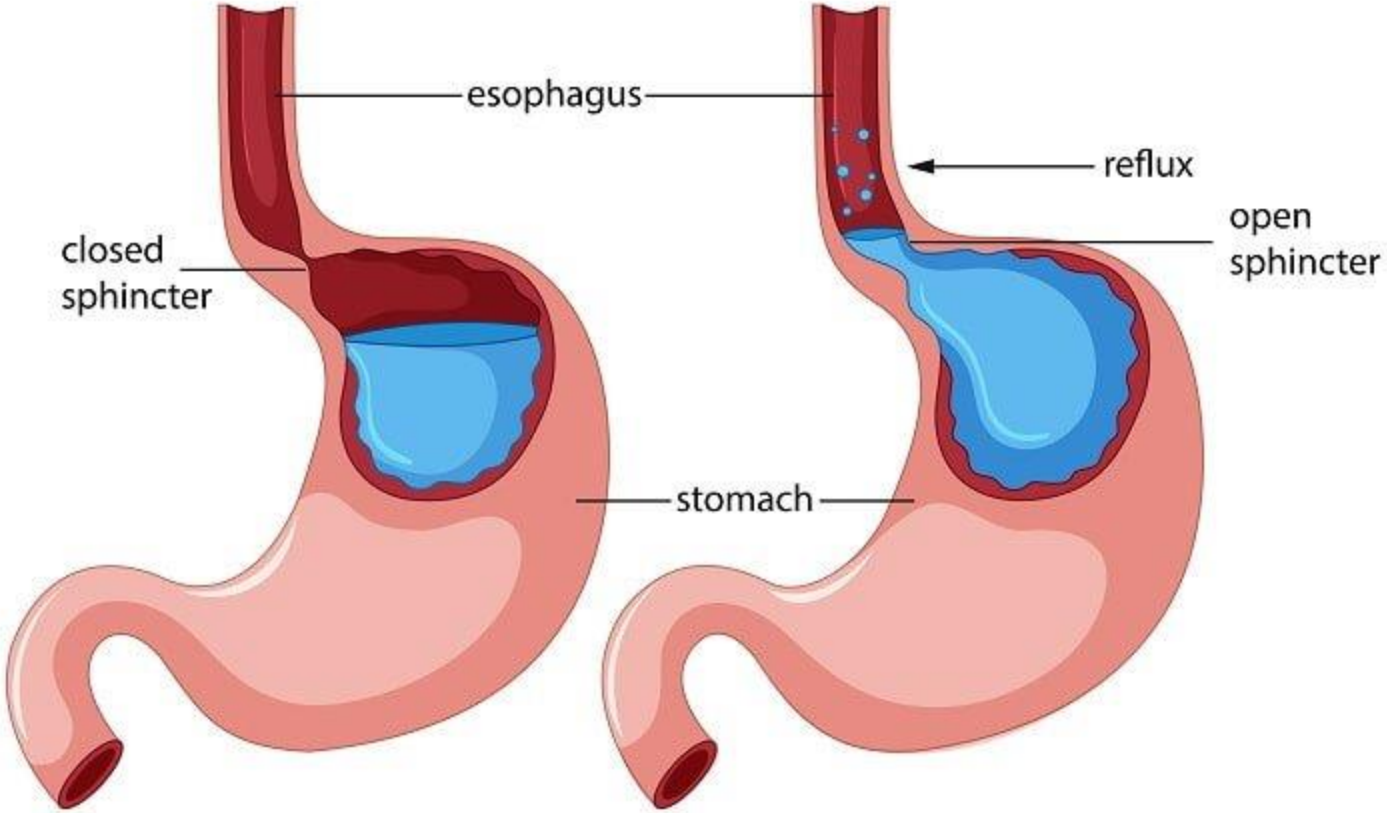
Gastroesophageal Reflux Disease

- Gastroesophageal reflux disease (GERD) is a disorder in which contents of the stomach or small intestine repeatedly move back up into the esophagus. This regurgitation action is called reflux.
- Reflux causes heartburn, and although most people have an attack of heartburn at some point in their lives, persistent reflux and severe heartburn may indicate GERD.

Gastroesophageal Reflux Disease

- Normally when a person swallows, the muscular walls of the esophagus move in waves to push the food down into the stomach.
- Right after the food enters the stomach, a band of muscle (the lower esophageal sphincter) closes off the esophagus.
- If the muscle fails to close tightly enough, the stomach contents can back up into the esophagus causing heartburn and other symptoms of GERD.

Gastroesophageal reflux disease (GERD)



Substances in the digestive juices from the stomach (namely, acid, pepsin, and bile) can damage the inner lining of the esophagus, causing inflammation ulcers (open sores), and, possibly, precancerous changes to cells.

Signs and Symptoms

1. Heartburn often occurs after a meal and worsens at night, when you are lying flat.
2. Belching Nausea and vomiting
3. Chronic cough, wheezing
4. Sore throat, hoarseness
5. change in voice, difficulty swallowing
6. Chest pain

Causes

- GERD can often be traced to the themalfunctioning of the lower esophageal sphincter (LES).
- The LES is a muscle responsible for opening and closing the lower end of the esophagus, which it accomplishes by relaxing and contracting at appropriate times.
- The LES may weaken or lose tone, allowing the contents of the stomach to escape back up the esophagus.

Any of the following may compromise the LES:

1.Pregnancy (in third trimester)

2.Obesity

3.Hiatal hernia (a condition in which a portion of the stomach protrudes through an opening in the diaphragm where the esophagus normally fits snuggly and passes into the stomach.)

4.Recurrent or persistent vomiting

5.Nasogastric tubes (these are inserted through the nose and into the stomach for a variety of medical reasons)

6.Smoking cigarettes

7.Alcohol intake

8.Psychological stress 9.Prior esophageal surgery

10.Esophageal stricture (narrowing of the esophagus)

11.Some medications including calcium channel blockers (a class of meds used for high blood pressure), anticholinergic drugs, iron pills, non-steroidal anti-inflammatory potassium, drugs, dopamine, sedatives and beta blockers

Treatment Approach

1- Lifestyle

1.Avoid behavior that does not allow food to easily move down into and through the stomach. This includes bending, lying down, or participating in exercises soon after a meal.

2.Don't eat heavy meals

3.Avoid acidic foods and drinks. These include caffeinated beverages, decaffeinated coffee, and orange juice

4. Avoid alcohol, chocolate, spearmint, and peppermint. These can relax the low esophageal sphincter.
5. Avoid carbonated beverages
6. Avoid eating fatty foods, including full-fat milk, which also may relax the lower esophageal sphincter tone.
7. Lose weight if you are overweight
8. Quit smoking
9. Avoid medications that provoke symptoms

If you experience more frequent symptoms at night, you may be relieved by:

1. Raising the head of your bed about six inches
2. Sleep on left side.
3. **Overcome stress.**

2- Medications

The main aim of drug treatment is to control the acidity of the stomach by the following:

1. **Antacids:**

typically recommended for mild symptoms of GERD; mostly a combination of three active ingredients: magnesium salts (often combined with aluminum to prevent diarrhea); calcium carbonate; and aluminum salts.

1. **Histamine H2 receptor blockers (such as cimetidine, and ranitidine): relieve symptoms for 6 to 24 hours; usually taken at bedtime.**

2- Medications

3.Proton pump inhibitors (including esomeprazole, lansoprazole, omeprazole, pantoprazole, and rabeprazole): are the most effective class of drugs for relieving symptoms of GERD; these drugs also speed the healing of the esophagus.

3.Prokinetic agent : specifically metoclopramide, which promotes movement of stomach acids along the gastrointestinal tract, rather than backing up into the esophagus.

4- Nutrition and Dietary Supplements

Bromelain (Ananas comosus)

Bromelain is an enzyme found in the pineapple plant. It contains protein-

digesting enzymes, which promote and maintain proper digestion and relieve symptoms of stomach upset or heartburn, particularly when used with other enzymes such as amylase (which digests starch) and lipase (which digests fat).

i.e. Bromelain + Amylase + Lipase = Proper digestion.

Selenium

Selenium, which is found in large quantities in whole grain products, may have some protection against precancerous cellular changes along the lining of the esophagus that can result from GERD.

5- Herbs

Liquorice (*Glycyrrhiza glabra*)

1.Licorice is a demulcent (soothing, coating agent) often used to prevent and treat stomach ulcers.

2.Preparations containing only glycyrrhizin (an active compound in licorice) are not so effective as anti-ulcer medications.

3.Liquorice extract (glycyrrhizin + flavonoids) was found to be the most effective in relieving pain associated with stomach ulcers and preventing the ulcers from recurring.

Turmeric (*Curcuma longa*)

❖Turmeric has long been used in both Ayurvedic and Chinese medicine to treat digestive disorders.

❖Extracts of turmeric root reduced the release of acid from the stomach and protected against injuries such as inflammation and ulcers, both of which are potential complications from GERD.

❖Note: at very high doses, turmeric may induce ulcers. So must be used in small doses

Astragalus (*Astragalus membranaceus*) Clinically, astragalus is used mainly for disorders of the digestive system, including ulcers.

Warnings and Precautions

People with high blood pressure, obesity, diabetes should avoid licorice. This herb should also not be used by pregnant or breastfeeding women or by men with decreased libido or other sexual dysfunctions. Use of any licorice product is not recommended for longer than four to four weeks.

Complications

The acidic contents of the stomach can damage the esophagus, causing narrowing, ulcers, erosion , and precancerous changes to cells.